

$$\therefore x + 8 = \frac{9}{7}x \Rightarrow 7x + 56 = 9x$$

$$\Rightarrow 2x = 56$$

$$\Rightarrow x = 28.$$

$\therefore$ Farah's age now = $(x + 8)$ years = $(28 + 8)$ years = 36 years.

Her daughter's age now = $\left(\frac{1}{6} \times 36\right)$ years = 6 years.

Her daughter's age 3 years ago = $(6 - 3)$ years = 3 years.

8. Let the daughter's age be x years. Then, mother's age = $3x$ years.

$$3x + 12 = 2(x + 12)$$

$$\Rightarrow 3x + 12 = 2x + 24$$

$$\Rightarrow x = 12.$$

Present age of daughter = 12 years.

9. Let the son's age be x years. Then, Mr. Sanyal's age = $3x$ years.

$$\therefore \frac{3x+6}{x+6} = \frac{5}{2} \Rightarrow 2(3x+6) = 5(x+6)$$

$$\Rightarrow 6x + 12 = 5x + 30$$

$$\Rightarrow x = 18.$$

$\therefore$ Present age of Mr. Sanyal = $3x$ years = (3×18) years = 54 years.

10. Average age of man and his daughter = 34 years.

Their total age = (34×2) years = 68 years.

Let man's age be x years. Then, daughter's age = $(68 - x)$ years.

$$\therefore \frac{x+4}{68-x+4} = \frac{14}{5} \Rightarrow 5(x+4) = 14(72-x)$$

$$\Rightarrow 5x + 20 = 1008 - 14x$$

$$\Rightarrow 19x = 988$$

$$\Rightarrow x = 52.$$

$\therefore$ Daughter's present age = $(68 - 52)$ years = 16 years.

11. Rani's age : Komal's age = $3 : 5 = \frac{3}{5} : 1$.

$$\text{Komal's age : Pooja's age} = 2 : 3 = 1 : \frac{3}{2}.$$

$$\text{Rani's age : Komal's age : Pooja's age} = \frac{3}{5} : 1 : \frac{3}{2} = 6 : 10 : 15.$$

Let Rani's age be $6x$ years. Then, Komal's age = $10x$ years and Pooja's age = $15x$ years.

$$\text{Rani's age} = \frac{2}{5} \text{ of Pooja's age} \Rightarrow 6x = \frac{2}{5} \times 15x.$$

Thus, we can not find the value of x and therefore of $6x$.

So, the answer cannot be determined.

12. Let son's age 10 years ago be x years.

Then, father's age 10 years ago = $3x$ years.

Son's age now = $(x + 10)$ years, Father's age now = $(3x + 10)$ years.

$$(3x + 10) + 10 = 2[(x + 10) + 10]$$

$$\Rightarrow 3x + 20 = 2(x + 20)$$

$$\Rightarrow 3x + 20 = 2x + 40$$

$$\Rightarrow x = 20.$$

Ratio of present ages of father and son

$$= \frac{3x+10}{x+10} = \frac{3 \times 20 + 10}{20 + 10} = \frac{70}{30} = \frac{7}{3} = 7 : 3.$$

13. Ram's age : Mohan's age = $4 : 5 = 1 : \frac{5}{4}$.

$$\text{Ram's age : Anil's age} = 5 : 6 = 1 : \frac{6}{5}.$$

Let Ram's age be x years. Then, Mohan's age = $\frac{5x}{4}$ years.

And, Anil's age = $\frac{6x}{5}$ years.

$$\therefore x + \frac{5x}{4} + \frac{6x}{5} = 69$$

$$\Rightarrow 20x + 25x + 24x = 1380$$

$$\Rightarrow 69x = 1380$$

$$\Rightarrow x = 20.$$

$$\text{Mohan's age} = \frac{5x}{4} \text{ years} = \frac{5 \times 20}{4} \text{ years} = 25 \text{ years.}$$

14. Let daughter's age be x years. Suresh's age = $2x$ years.

$$\therefore \frac{2x+6}{x+6} = \frac{23}{13} \Rightarrow 13(2x+6) = 23(x+6)$$

$$\Rightarrow 26x + 78 = 23x + 138$$

$$\Rightarrow 3x = 60$$

$$\Rightarrow x = 20.$$

Present age of Suresh = $2x$ years

$$= (2 \times 20) \text{ years} = 40 \text{ years.}$$

15. Let the ages of Arun and Deepak 7 years ago be $5x$ years and $7x$ years respectively. Then,

Arun's present age = $(5x + 7)$ years, Deepak's present age = $(7x + 7)$ years.

$$\therefore (7x + 7) - (5x + 7) = 14$$

$$\Rightarrow 2x = 14$$

$$\Rightarrow x = 7.$$

Deepak's present age = $(7 \times 7 + 7)$ years = 56 years.

16. Let son's age 10 years ago be x years. Then, man's age 10 years ago = $7x$ years.

Son's present age = $(x + 10)$ years, Man's present age = $(7x + 10)$ years.

$$\therefore 2[(7x + 10) + 2] = 5[(x + 10) + 2]$$

$$\Rightarrow 2(7x + 12) = 5(x + 12)$$

$$\Rightarrow 14x + 24 = 5x + 60$$

$$\Rightarrow 9x = 36$$

$$\Rightarrow x = 4.$$

$\therefore$ Son's present age = $(x + 10)$ years = $(4 + 10)$ years = 14 years.

17. Let Samina's age be $7x$ years. Then, Suhana's age = $3x$ years.

$$\therefore \frac{7x+6}{3x+6} = \frac{5}{3} \Rightarrow 3(7x+6) = 5(3x+6)$$

$$\Rightarrow 21x + 18 = 15x + 30$$